

Healthcare Patient Management System

From a flat CSV to a secured, audited, scheduled data system

UCI Diabetes 130-US Hospitals, 1999-2008 | Educational use only

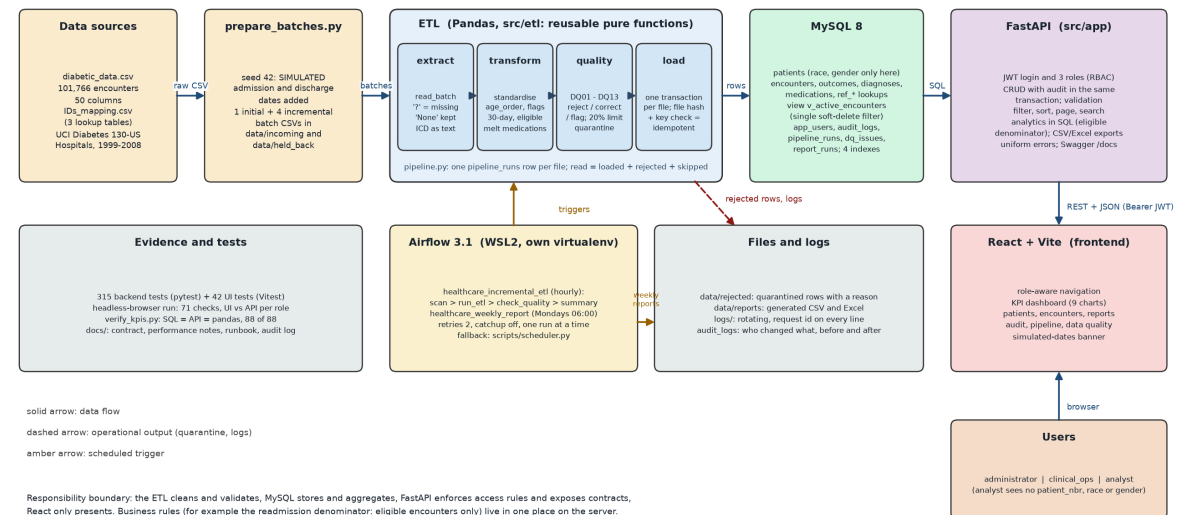
Dates in this data are simulated | Findings are associations, not causes

The problem and four questions

- 101,766 encounters, 71,518 patients, flat CSV, no dates
- Q1 What is the 30-day readmission rate, and for whom?
- Q2 How long do patients stay?
- Q3 How do admissions trend over time? (simulated dates)
- Q4 Which medication, lab and prior-visit patterns are associated with readmission?

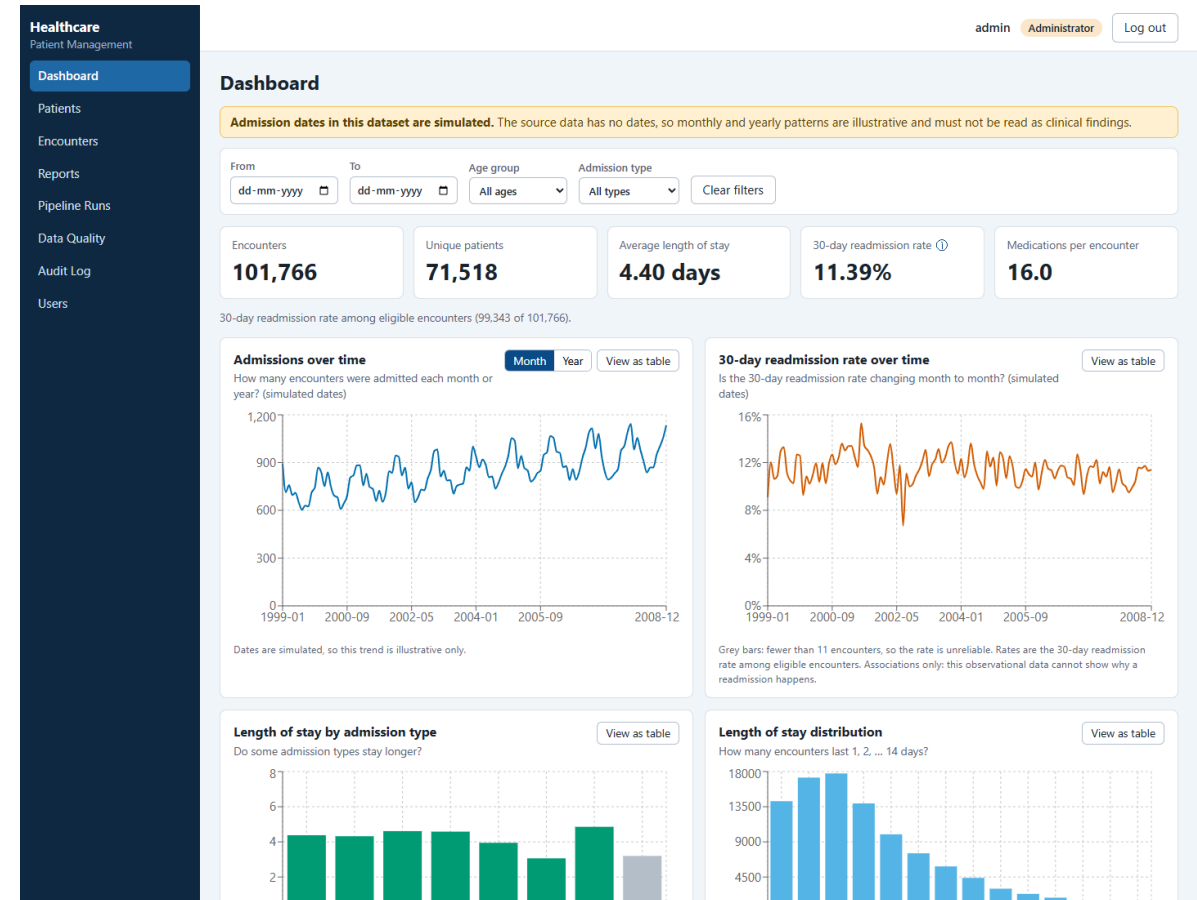
Healthcare Patient Management System - Architecture

Dataset -> ETL -> MySQL -> FastAPI -> React, scheduled by Airflow. Everything runs locally (no cloud, Docker or CI/CD). Admission and discharge dates are SIMULATED.



Solution overview

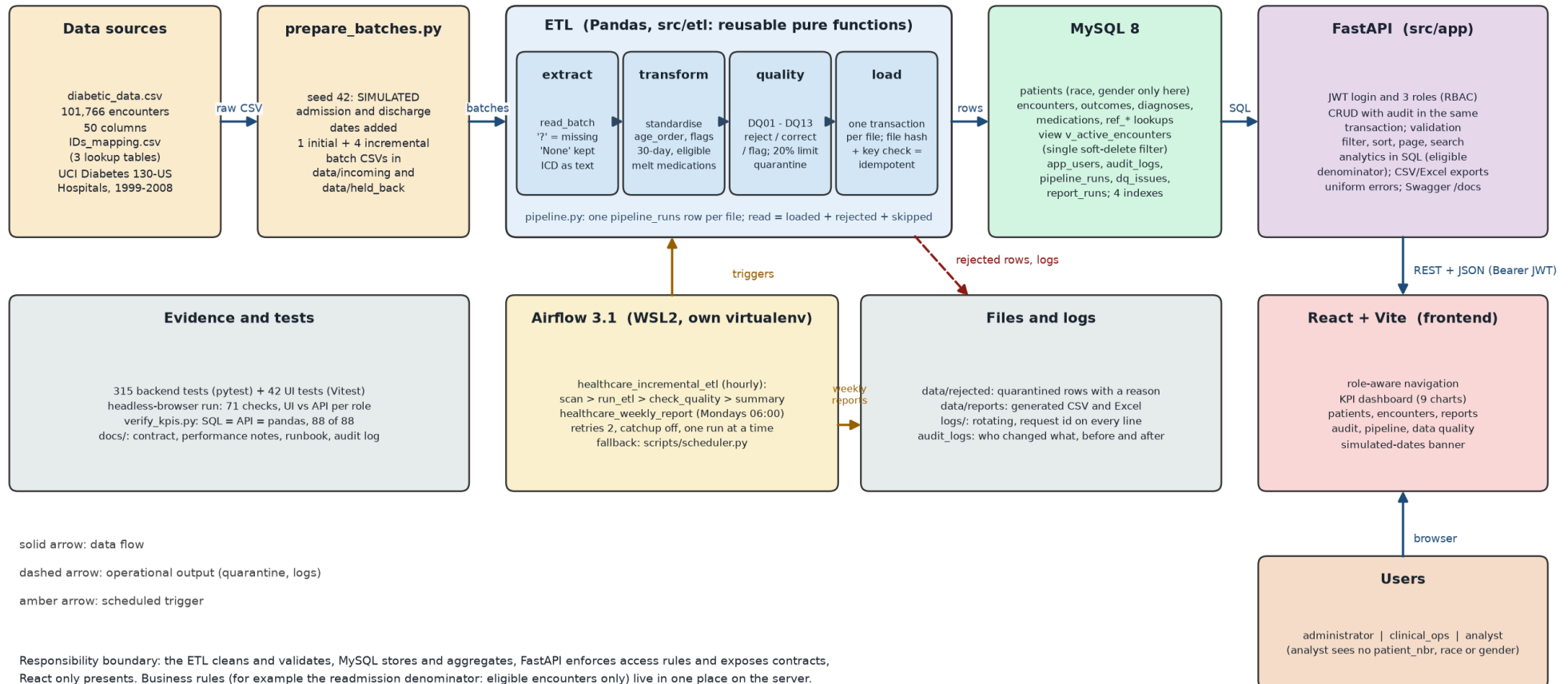
- ETL: 13 quality rules, quarantine, idempotent reloads
- MySQL: 16 tables, one view as the soft-delete filter
- FastAPI: 32 operations, JWT, 3 roles, audit trail
- React dashboard, CSV and Excel reports
- Airflow: hourly load, weekly reports



Architecture and data flow

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Dataset to ETL to MySQL to API to React; Airflow triggers; quarantine, logs and reports on the side

Data and quality rules

- '?' is missing; the word 'None' is a real lab value
- DQ01-DQ13: reject, correct, flag, measure
- Bad file (over 20% rejected) loads nothing
- Faulty batch: 23 rows quarantined, with reasons
- Same file twice: SKIPPED_DUPLICATE_FILE

Healthcare
Patient Management

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Audit Log

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admin Administrator Log out

Data quality

Row-level findings from each load. "rejected" rows were quarantined, "corrected" values were set to empty, "flagged" rows were kept, "metric" rows count missing values.

Run

Rule

Severity

1

DQ03

Any

Apply

Clear

ISSUE ▲	RUN	RULE	SEVERITY	ACTION	ENCOUNTER	COLUMN	VALUE
1	1	DQ12	Info	flagged	98634858	gender	Unknown/Invalid
2	1	DQ12	Info	flagged	226864668	gender	Unknown/Invalid
3	1	DQ13	warning	flagged	3301014	race	Caucasian
4	1	DQ13	warning	flagged	4066842	race	Hispanic
5	1	DQ13	warning	flagged	7497384	race	Caucasian
6	1	DQ13	warning	flagged	8373876	race	Caucasian
7	1	DQ13	warning	flagged	10943376	race	Caucasian
8	1	DQ13	warning	flagged	11093766	race	Other
9	1	DQ13	warning	flagged	12367974	race	Caucasian
10	1	DQ13	warning	flagged	12385578	race	Other
11	1	DQ13	warning	flagged	14824488	race	Caucasian
12	1	DQ13	warning	flagged	15876660	race	Caucasian
13	1	DQ13	warning	flagged	20729328	gender	Female
14	1	DQ13	warning	flagged	20884452	race	Caucasian
15	1	DQ13	warning	flagged	21486336	race	AfricanAmerican
16	1	DQ13	warning	flagged	21937122	race	Hispanic
17	1	DQ13	warning	flagged	23766162	race	Caucasian
18	1	DQ13	warning	flagged	24623514	race	Caucasian
19	1	DQ13	warning	flagged	27055116	race	Other
20	1	DQ13	warning	flagged	29459682	race	Other
21	1	DQ13	warning	flagged	33960042	race	Other
22	1	DQ13	warning	flagged	36462060	race	Other
23	1	DQ13	warning	flagged	36685962	race	Caucasian
24	1	DQ13	warning	flagged	38055744	race	Caucasian
25	1	DQ13	warning	flagged	38721822	race	AfricanAmerican

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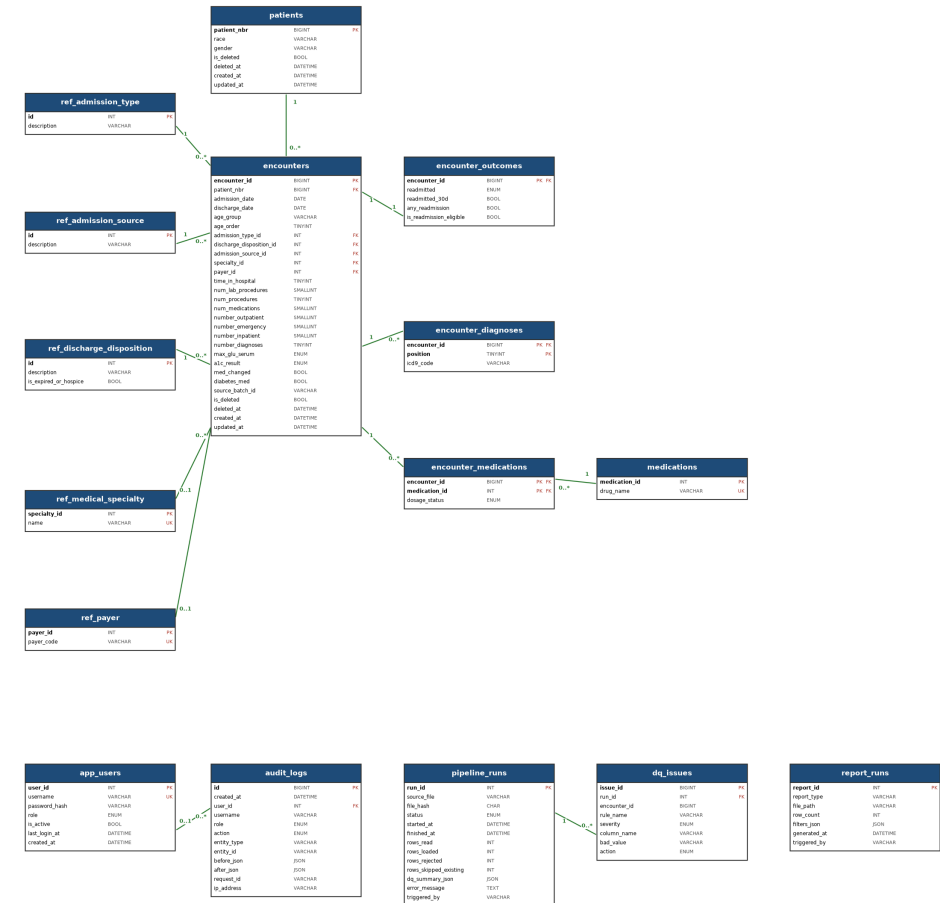
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Database design

- Third normal form: race and gender on patients only
- Diagnoses and medications as child tables
- Medications: only prescribed rows, 120,054 vs 2.1M
- 4 indexes, each proven by EXPLAIN
- Date-range list: 256 ms to 0.8 ms; ICD search: 765 ms to 9 ms

Healthcare Patient Management System - ERD (admission/discharge dates are SIMULATED)

PK = primary key, FK = foreign key, UK = unique, Keys encounter_id and patient_id are BIGINT from the source (not auto-generated), race/gender exist only on patients.



API and security

- JWT, 60 minutes; role read from the database each request
- Roles enforced on the server, not in the UI
- Analyst: no patient id, race or gender; 403 on patients
- Audit row written in the same transaction as the change
- Errors: one contract, never a stack trace

The screenshot shows the 'Healthcare Patient Management' application. On the left is a dark sidebar with navigation links: Dashboard, Patients, Encounters, Reports, Pipeline Runs, Data Quality, **Audit Log**, and Users. The main content area is titled 'Audit log' and includes a sub-header: 'Every create, update, delete and encounter-level export, with the values before and after. Passwords and tokens are never recorded.'

At the top right of the main area, there is a user profile for 'admin' with the role 'Administrator' and a 'Log out' button.

Below the header is a search and filter section with fields for 'User', 'Action' (set to 'Any'), 'Entity' (set to 'patient, encounter...'), 'Entity ID' (set to '910845266'), 'From' (date), and 'To' (date). There are 'Apply' and 'Clear' buttons.

The main part of the interface is a table with the following columns: WHEN ▲, USER, ACTION, ENTITY, ID, and DETAILS. It contains three rows of audit data:

WHEN ▲	USER	ACTION	ENTITY	ID	DETAILS
2026-10-09 21:20	admin (administrator)	CREATE	encounter	910845266	Show before/after
2026-10-09 21:20	admin (administrator)	UPDATE	encounter	910845266	Hide
2026-10-09 21:20	admin (administrator)	DELETE	encounter	910845266	Show before/after

Below the table, there is a section for a specific request: 'Request f2dbdf1c1962466e · IP 127.0.0.1'. It contains a table with columns 'FIELD', 'BEFORE', and 'AFTER':

FIELD	BEFORE	AFTER
readmitted	"NO"	"<30"
readmitted_30d	false	true
any_readmission	false	true

At the bottom of the main area, there is a pagination indicator '1-3 of 3' and navigation buttons: 'Previous', 'Page 1 of 1', and 'Next'.

At the very bottom of the sidebar, there is a small disclaimer: 'Educational use only. De-identified research data, 1999-2008.'

Dashboard results

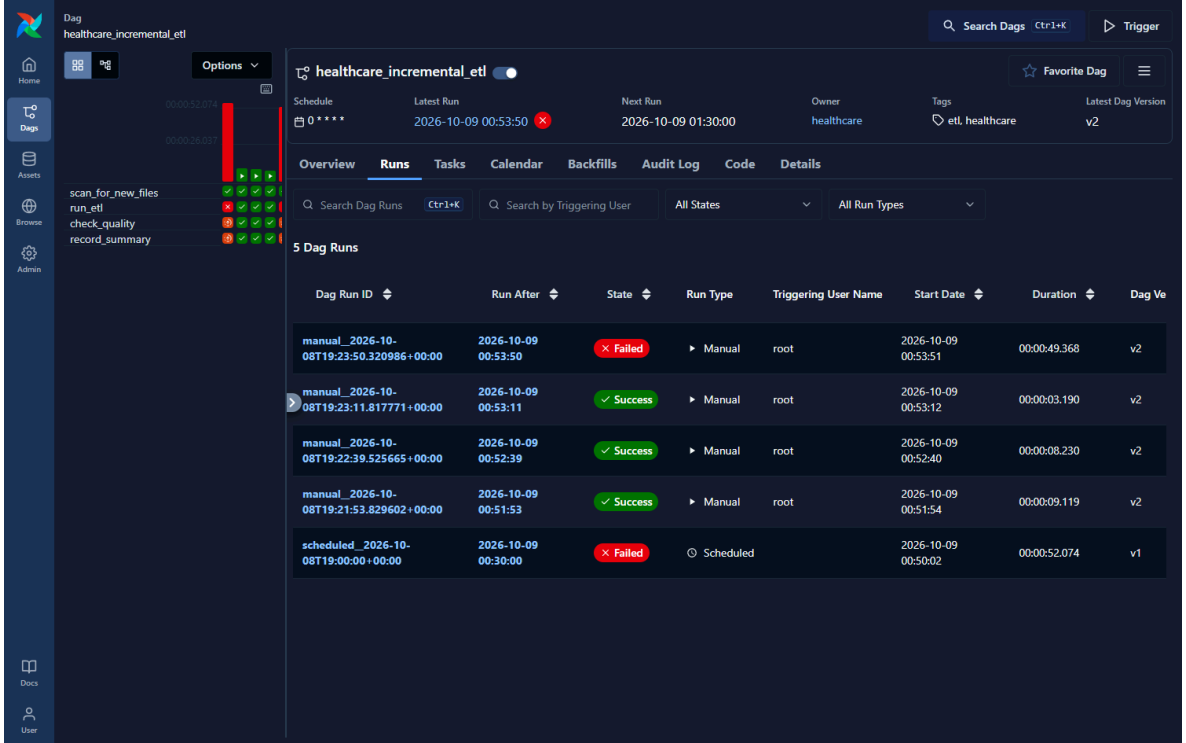
- Prior inpatient visits 0 vs 3+: 8.6% vs 26.4% readmitted within 30 days
- Insulin dose 'Down' 14.2% vs 'No insulin' 10.2%



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Testing and reliability

- 315 backend tests on a real MySQL schema, 94% line coverage
- 42 UI tests; 71 real-browser checks per role
- 88 KPI checks: SQL = API = pandas on the raw CSV
- Deliberate bug (wrong denominator): 6 tests fail
- Hard-kill test: nothing half-loaded



healthcare_incremental_etl

Schedule: 0 ****, Latest Run: 2026-10-09 00:53:50 (Failed), Next Run: 2026-10-09 01:30:00, Owner: healthcare, Tags: ett, healthcare, Latest Dag Version: v2

Overview | **Runs** | Tasks | Calendar | Backfills | Audit Log | Code | Details

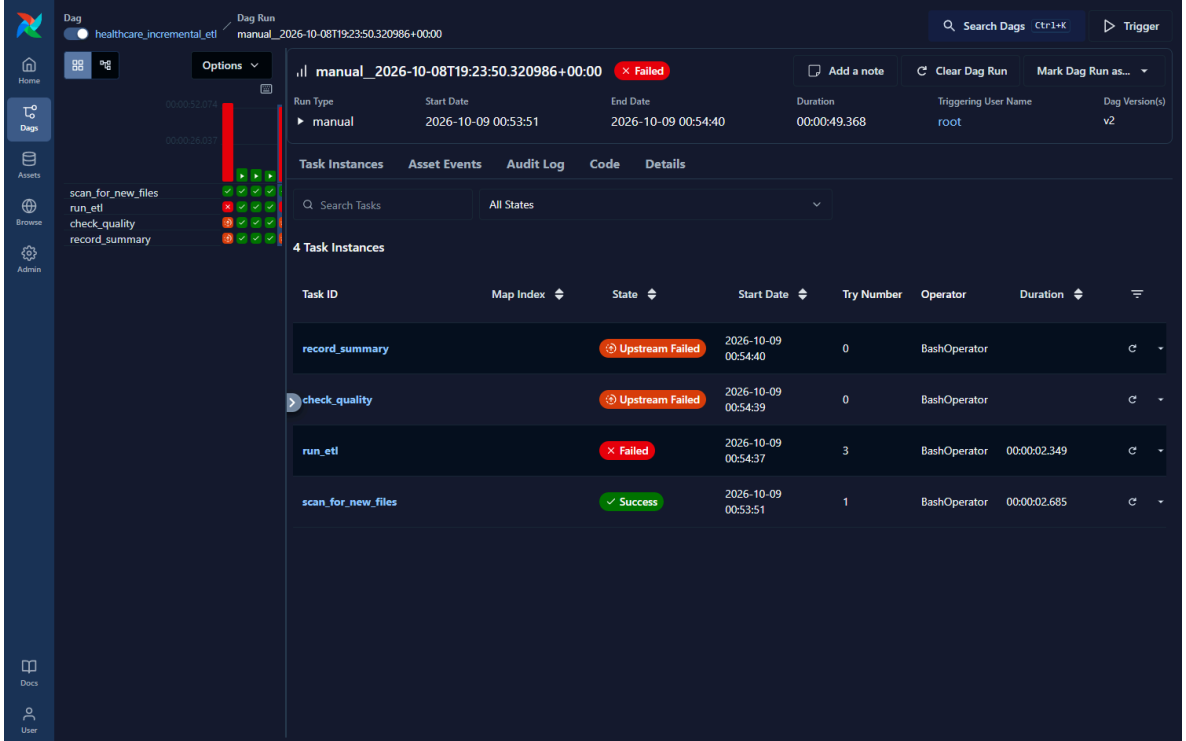
Search Dag Runs: Ctrl+K, Search by Triggering User: All States, All Run Types

5 Dag Runs

Dag Run ID	Run After	State	Run Type	Triggering User Name	Start Date	Duration	Dag Ver
manual_2026-10-08T19:23:50.320986+00:00	2026-10-09 00:53:50	Failed	Manual	root	2026-10-09 00:53:51	00:00:49.368	v2
manual_2026-10-08T19:23:11.817771+00:00	2026-10-09 00:53:11	Success	Manual	root	2026-10-09 00:53:12	00:00:03.190	v2
manual_2026-10-08T19:22:39.525665+00:00	2026-10-09 00:52:39	Success	Manual	root	2026-10-09 00:52:40	00:00:08.230	v2
manual_2026-10-08T19:21:53.829602+00:00	2026-10-09 00:51:53	Success	Manual	root	2026-10-09 00:51:54	00:00:09.119	v2
scheduled_2026-10-08T19:00:00+00:00	2026-10-09 00:30:00	Failed	Scheduled		2026-10-09 00:50:02	00:00:52.074	v1

Challenges, trade-offs and limits

- Simulated dates: trends are illustrative only
- No hospital id: categories compared, not hospitals
- Whole-file reject above 20%: safe, but strict
- No summary table: two queries stay at 0.6-0.8 s
- Airflow in WSL2: two environments to maintain



The screenshot displays the Apache Airflow web interface. On the left, a sidebar contains navigation links for Home, DAGs, Assets, Browse, and Admin. The main panel shows the details for a DAG named 'healthcare_incremental_etl'. The selected run is 'manual_2026-10-08T19:23:50.320986+00:00', which is marked as 'Failed'. Below this, a table lists the tasks and their execution status:

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
record_summary		Upstream Failed	2026-10-09 00:54:40	0	BashOperator	
check_quality		Upstream Failed	2026-10-09 00:54:39	0	BashOperator	
run_etl		Failed	2026-10-09 00:54:37	3	BashOperator	00:00:02.349
scan_for_new_files		Success	2026-10-09 00:53:51	1	BashOperator	00:00:02.685

Learnings and next steps

- A plausible number can be wrong: check the denominator
- Measure first; one index was rejected on evidence
- Run the commands: fresh-clone check found real gaps
- Next: real timestamps, hospital dimension
- Next: validated risk model; alerting on failed runs

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Pipeline runs

Each row is one file processed by the ETL. Read = loaded + rejected + skipped, always.

Status: Any From: dd-mm-yyyy To: dd-mm-yyyy Apply Clear

RUN	FILE	STATUS	STARTED	READ	LOADED	REJECTED	SKIPPED	REJECT RATIO	QUALITY
5	encounters_batch_004.csv	SUCCESS	2026-10-09 21:15	5,088	5,088	0	0	0.00%	Issues
4	encounters_batch_003.csv	SUCCESS	2026-10-09 21:15	5,088	5,088	0	0	0.00%	Issues
3	encounters_batch_002.csv	SUCCESS	2026-10-09 21:15	5,089	5,089	0	0	0.00%	Issues
2	encounters_batch_001.csv	SUCCESS	2026-10-09 21:15	5,089	5,089	0	0	0.00%	Issues
1	encounters_batch_000_initial.csv	SUCCESS	2026-10-09 21:15	81,412	81,412	0	0	0.00%	Issues

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